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Semester: 3rd

Department: BSCY

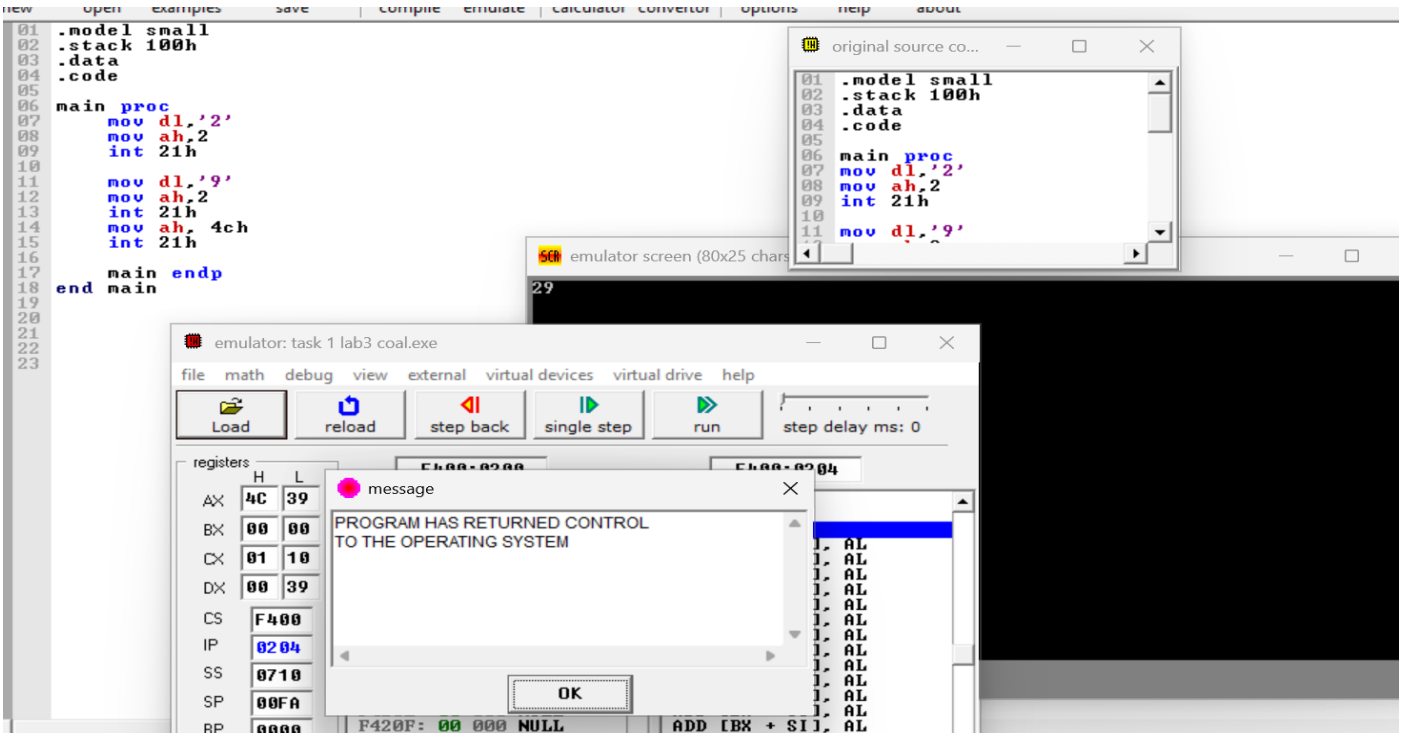
Course: COAL Lab

Instructor: Ma’am Rehana

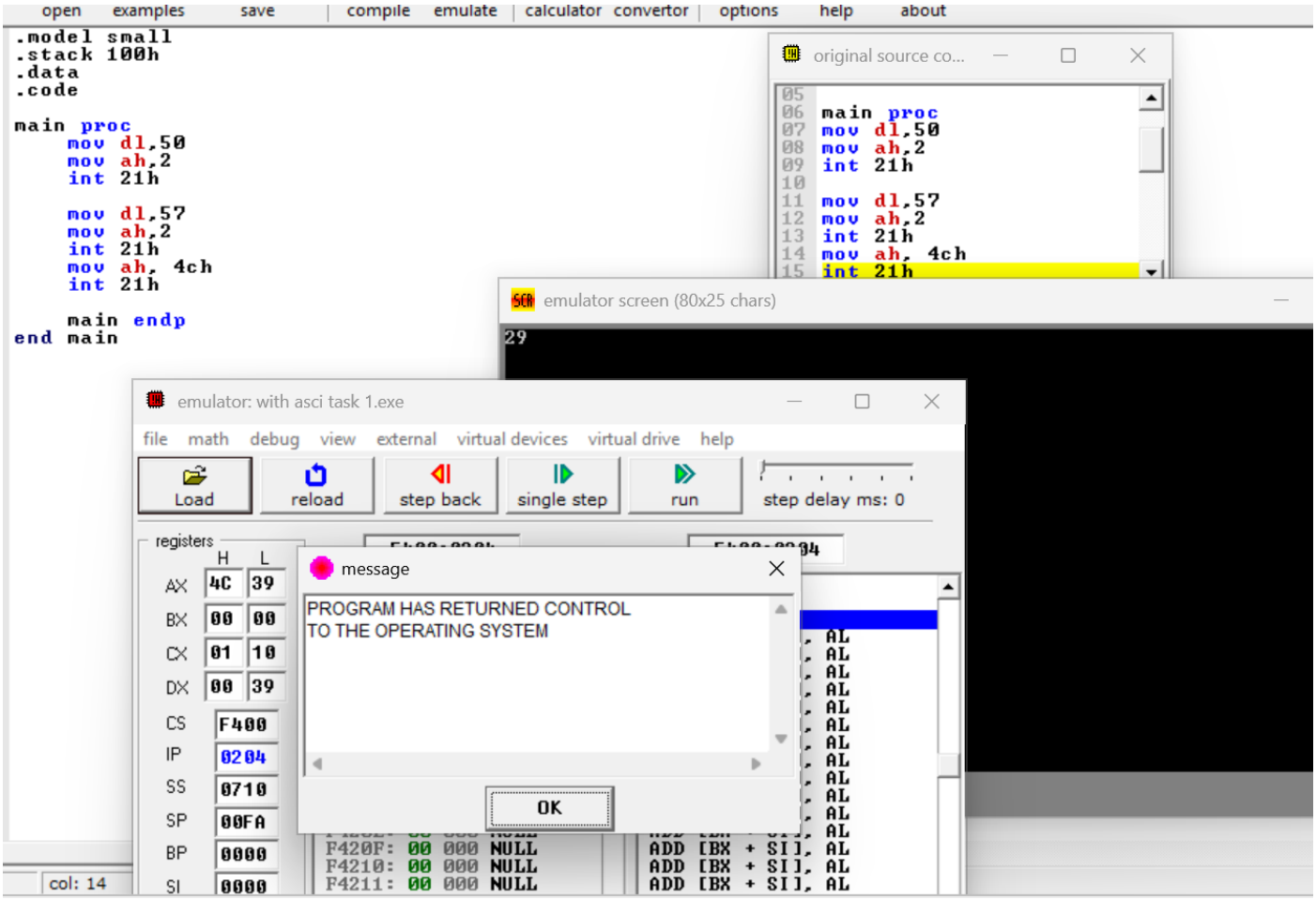
LAB 03

Q1. Write an assembly program to print a two numbers (2,9) on screen with and without ascii code?

Without ascii:



With ascii:



Q2. Write an assembly program to print a two characters (X,Y) on screen with and without ascii code?

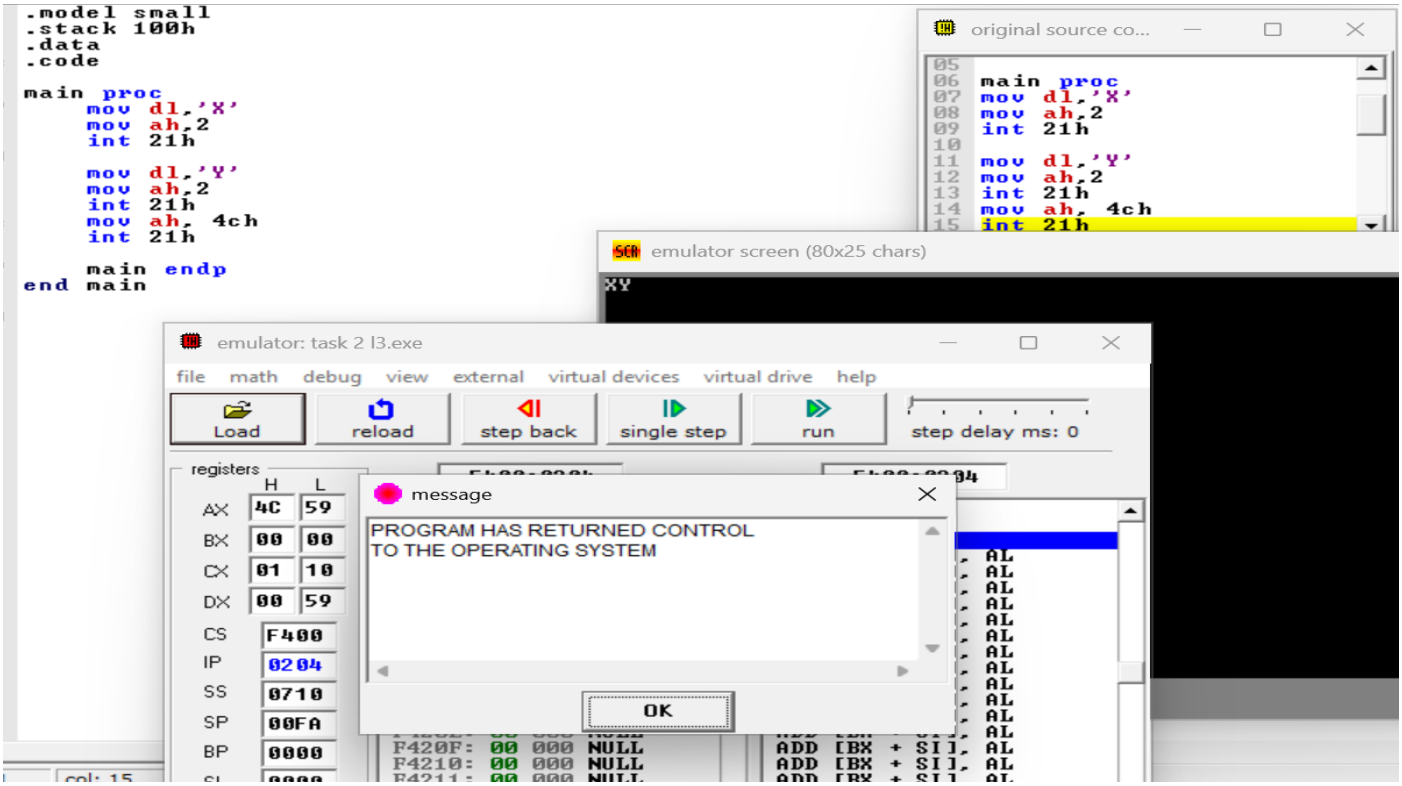
Without ascii:

```
.model small
.stack 100h
.data
.code

main proc
     mov dl,'X'
     mov ah,2
     int 21h

     mov dl,'Y'
     mov ah,2
     int 21h
     mov ah, 4ch
     int 21h

     main endp
end main
```



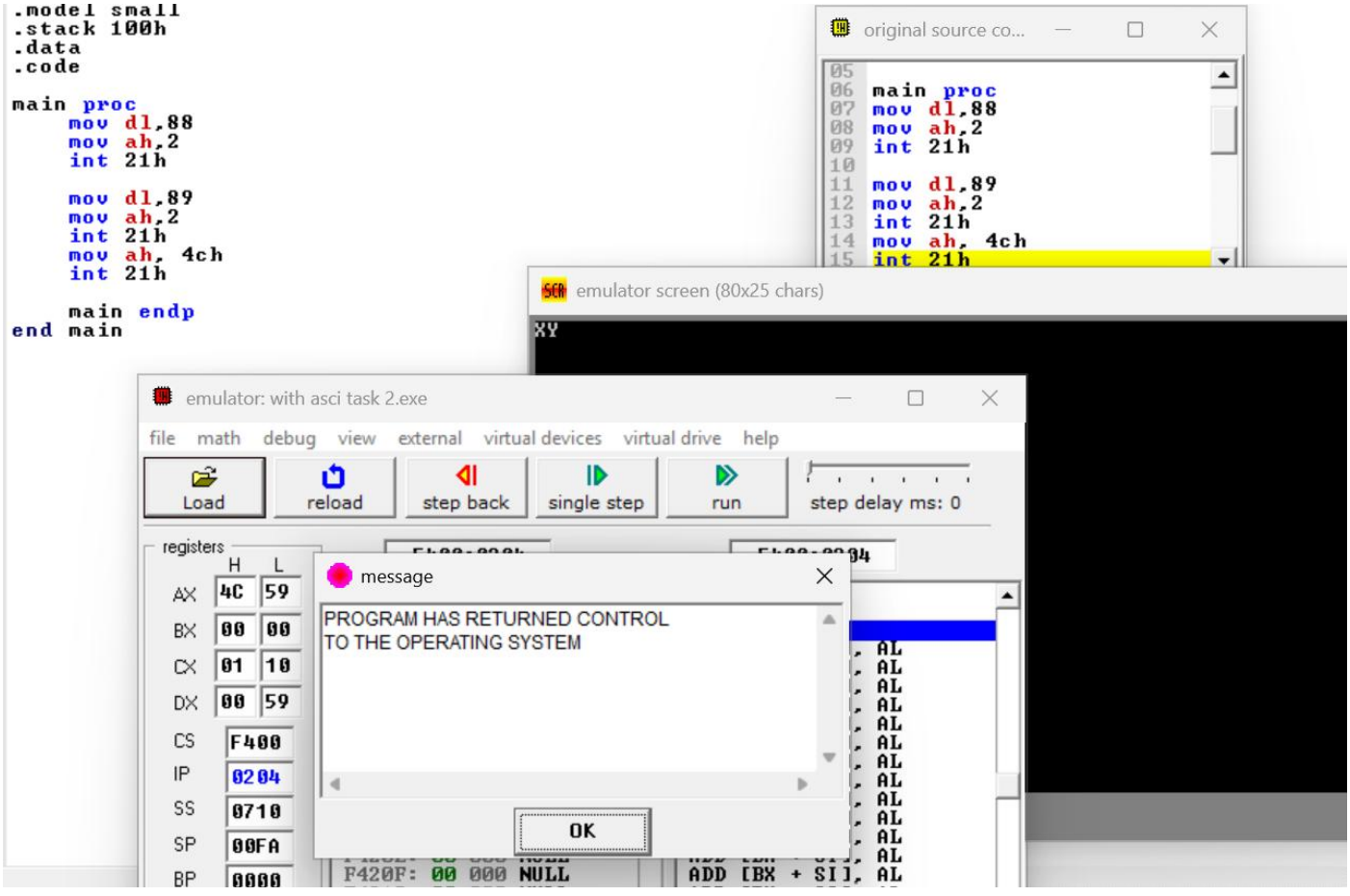
With ascii:

```
.model small
.stack 100h
.data
.code

main proc
     mov dl,88
     mov ah,2
     int 21h

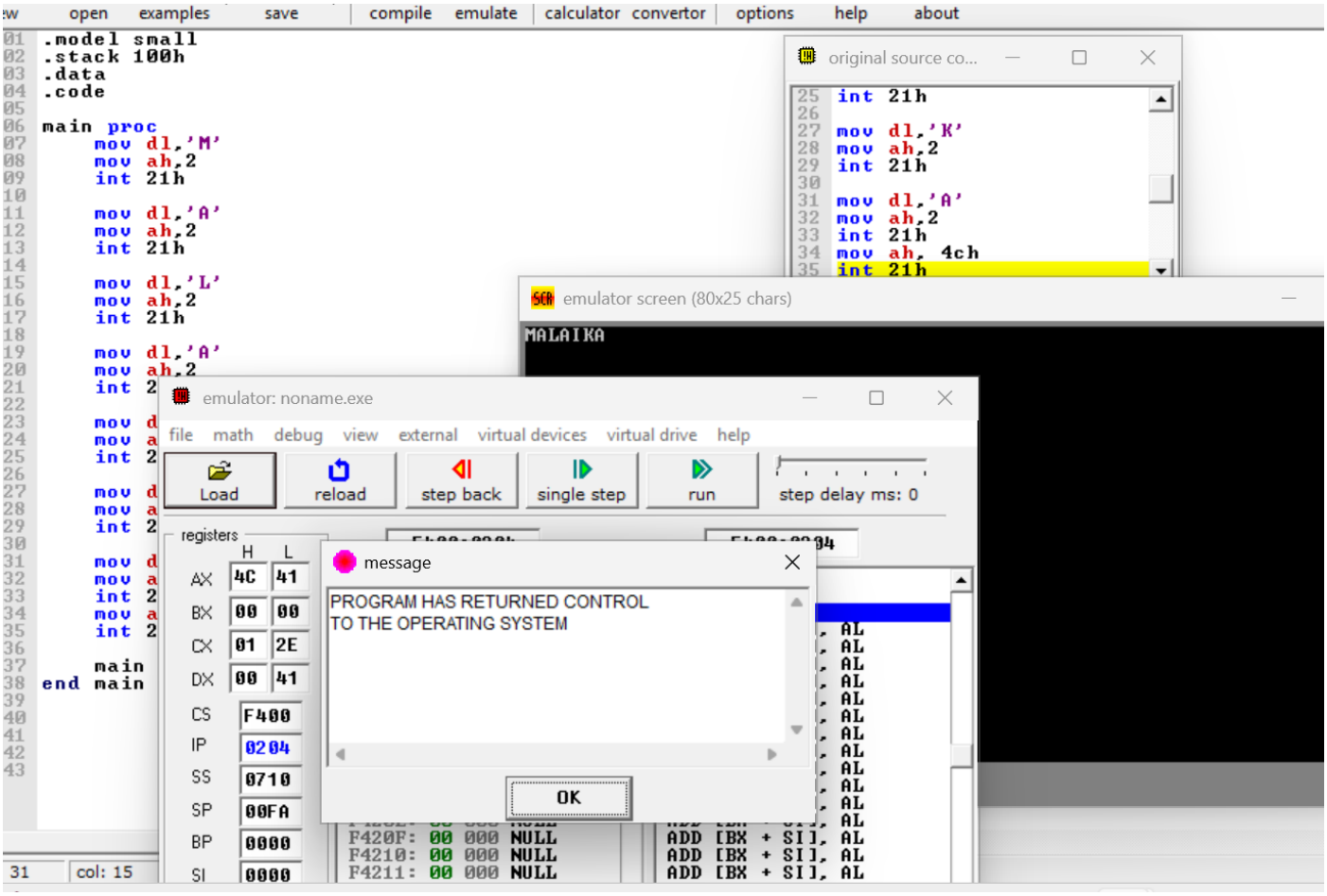
     mov dl,89
     mov ah,2
     int 21h
     mov ah, 4ch
     int 21h

     main endp
end main
```

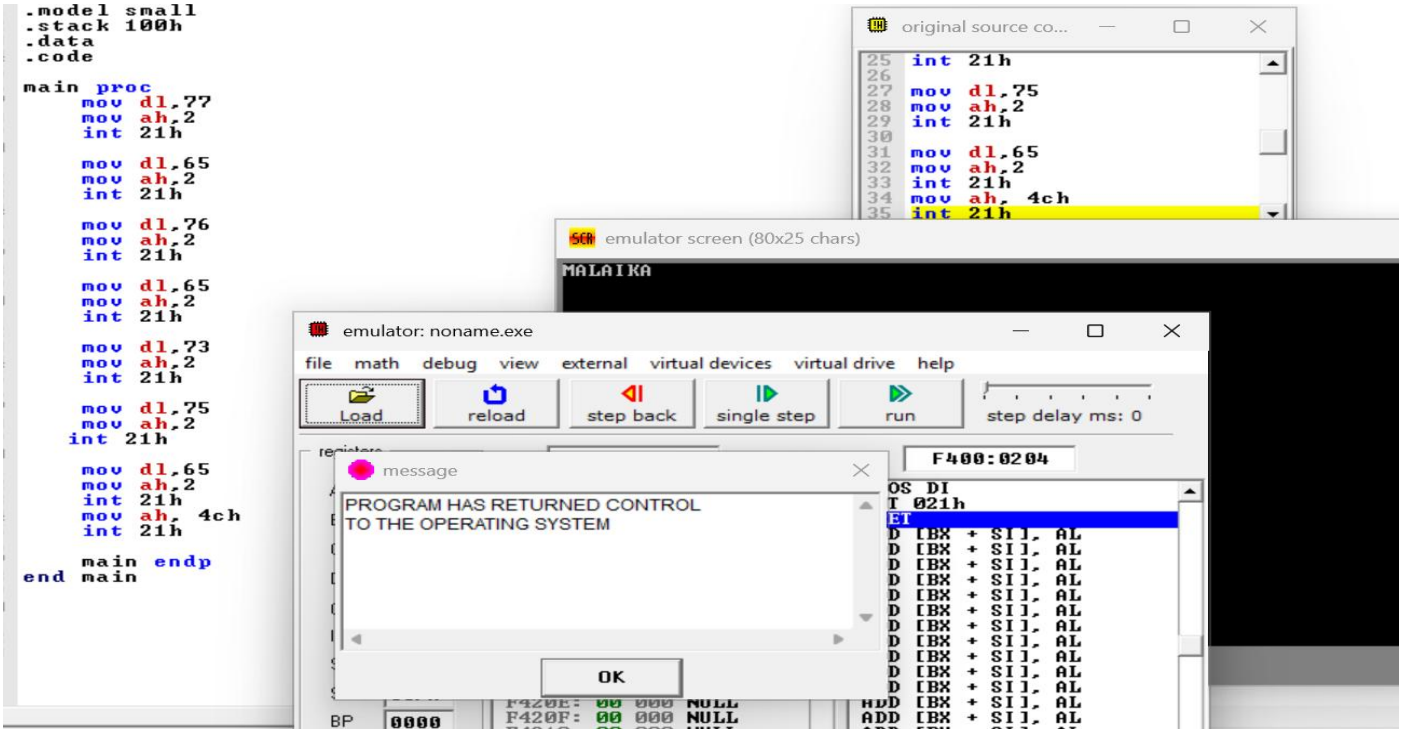


Q3. Print your name and sap id on screen?

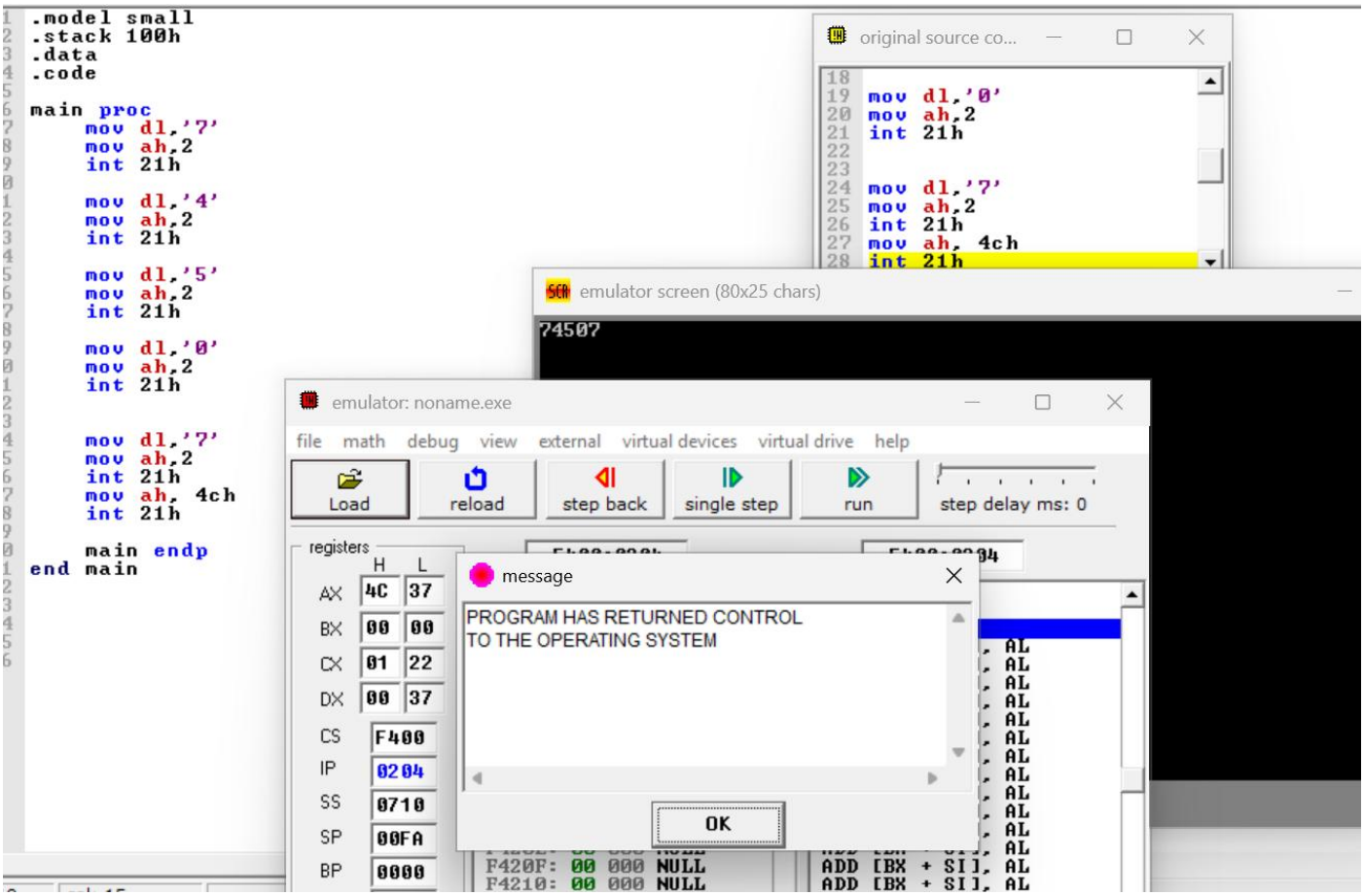
Without ascii:



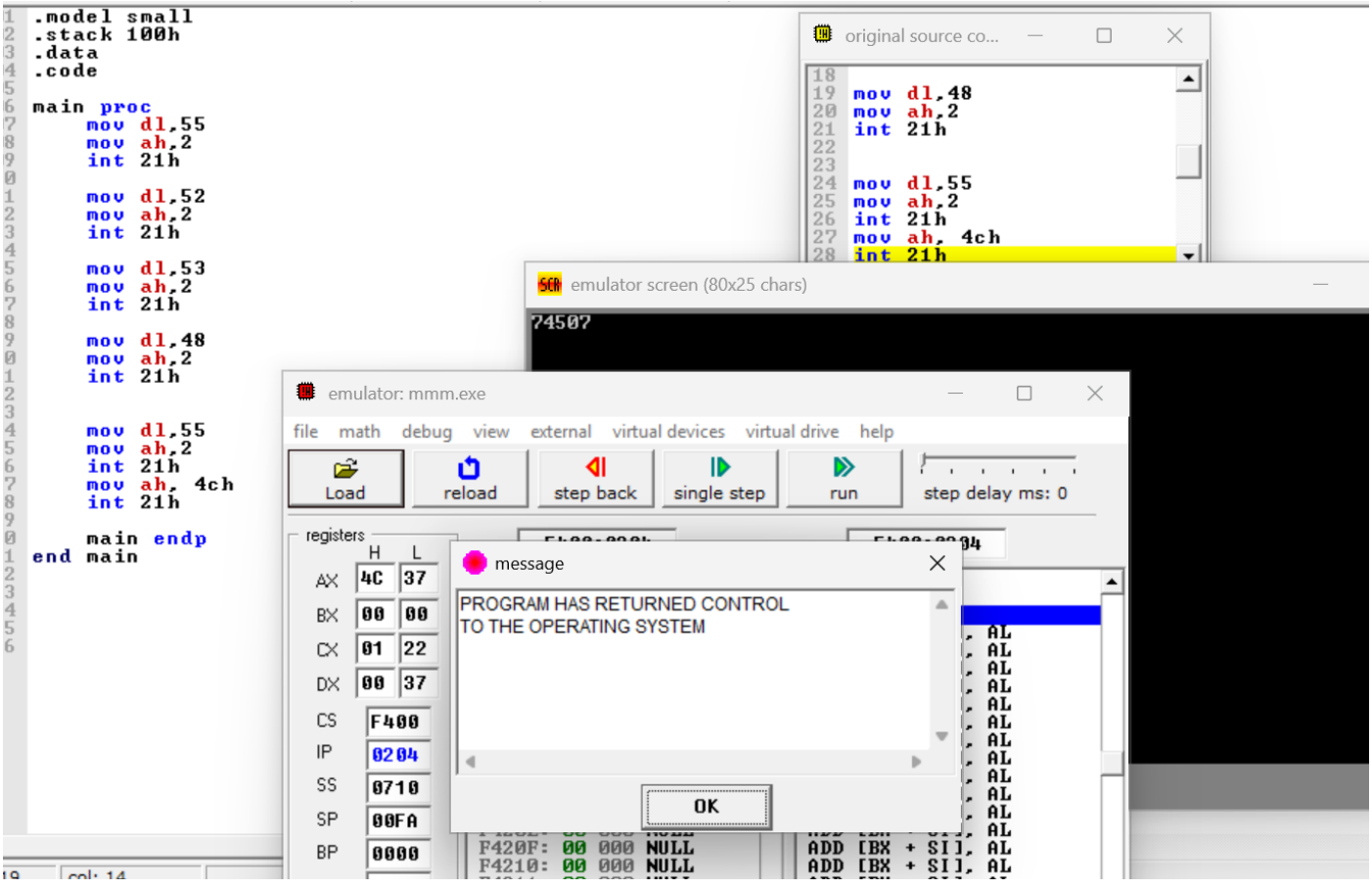
With ascii:



Without ascii:



With ascii:



Q4. Identify the addressing mode used in each of the following instructions.

MOV AL, 5
MOV AL, BL
MOV AL, [NUM]
MOV AL, [SI]

Ans: MOV AL, 5

- **Addressing Mode:** Immediate Addressing Mode
- **Reason:** The operand is a constant value (5) specified directly in the instruction.

MOV AL, BL

- **Addressing Mode:** Register Addressing Mode
- **Reason:** Both source (BL) and destination (AL) operands are CPU registers.

MOV AL, [NUM]

- **Addressing Mode:** Memory Addressing Mode
- **Reason:** The instruction accesses memory using a direct variable name/address ([NUM]).

MOV AL, [SI]

- **Addressing Mode:** Register Addressing Mode
- **Reason:** The effective memory address of the operand is held inside an index/pointer register (SI).

Q5. What are addressing mode explain each with example.

Ans: An **addressing mode** refers to the technique used by an instruction in assembly language to specify where the operand (data) is located whether in an immediate value, a CPU register, or a memory address.

- **Immediate Addressing Mode**
  - **Definition:** The operand data is supplied directly as part of the instruction itself.
  - **Example:** MOV AX, 100

- **Register Addressing Mode**
  - **Definition:** The operand is stored in a CPU register, and the operation takes place entirely within registers without accessing memory.

- **Example:** MOV CX, DX

- **Memory Addressing Mode**
  - **Definition:** The memory address containing the operand is explicitly specified inside the instruction.
  - **Example:** MOV AL, [2000H]